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June 20, 1975
SUPERSEDING
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FEDERAL SPECIFICATION

INK, WRITING, FOR FOUNTAIN AND DIP PENS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers writing inks for use in fountain and dip pens (see 6.1).

1.2 Classification.

1.2.1 Types and classes. Writing inks shall be of the following types and classes, as specified (see 6.2):

Type I - Washable fluid.

Type II - Permanent.
Class 1 - Fluid.
Class 2 - Powder.

1.2.2 Colors. Type I writing ink shall be blue. Type II writing ink shall be blue-black or jet black, as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

P-S-620 - Soap, Toilet, Cake (Floating and Nonfloating).
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-665 - Boxes: Paperboard, Metal Edged and Components.
PPP-B-676 - Boxes, Set-Up, Paperboard.
UU-P-121 - Paper, Bond and Writing, White and Colored.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

FSC 7510

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks, and the Index of Federal Specifications and Standards, from established distribution points in their agencies.)

Military Specification:

MIL-L-10547 - Liner, Case and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, NW., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material.

3.1.1 The material shall be suitable for the intended purpose as hereinafter specified. Ingredients shall be of the quality normally used for writing ink in standard practice. All ingredients shall be free from impurities that may affect the satisfactory storage and use of the finished product.

3.1.2 The ink shall contain no benzol (benzene), aniline oil, chlorinated compounds, or other toxic or hydrolyzable chlorine derivatives. A certificate of compliance with this requirement shall be furnished by the contractor.

3.2 Writing quality. When tested as described in 4.4.1, writing ink shall flow smoothly without clogging and shall show no evidence of feathering, smudging, or offsetting.

3.3 Stability. There shall be no mold, scum, deposit, or sediment when tested as described in 4.4.2.

3.4 Reaction. Writing ink shall remain stable when in contact with metal parts of fountain pen or dip set (see 4.4.3.1). The pen point used in the test in 4.4.3 shall not be grossly etched, corroded, or pitted so as to affect its writing qualities.

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3.5 Type I ink.

3.5.1 Washability. Type I ink, when tested as described in 4.4.4, shall not leave a stain on the test fabric.

3.6 Type II ink.

3.6.1 Light fastness. Writing shall be legible after being exposed as specified in 4.4.5.

3.6.2 Fastness to water. Writing, as described in 4.4.1, shall remain legible after being tested as specified in 4.4.6.

3.6.3 Class 2. Type II, class 2 powder, when dissolved in distilled water in the quantity specified on the label of the container, shall provide ink meeting the requirements for the type II ink of this specification.

3.7 Workmanship. The ink shall be smooth, homogeneous, free from lumps, strings, and objectionable odor. Labels shall be clearly printed and neatly and securely affixed to bottles and tubes. Ink containers shall show no evidence of spillage or leakage.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. For the purpose of sampling, an inspection lot for acceptance, inspection, and tests shall consist of all material and all bottles or tubes of ink of the same size, submitted for delivery at one time, unless otherwise indicated hereinafter.

4.2.1 Sampling and inspection for acceptance. Sampling and inspection shall be performed in accordance with the provisions of MIL-STD-105, except where otherwise indicated hereinafter.

4.2.2 Sampling for tests.

4.2.2.1 Fluid ink. An unopened bottle shall be taken from each lot, when such bottle contains one pint or more. When ink is furnished in smaller bottles, a quantity sufficient to aggregate at least one pint shall be taken. When more than one bottle is taken, no two bottles shall be taken from the same shipping container.

4.2.2.2 Type II, class 2. One sealed bottle or tube shall be taken from each lot to make at least one pint of fluid ink.

4.3 Inspection.

4.3.1 Inspection of materials and components. The supplier is responsible, in accordance with 4.1, for insuring that components and materials used were manufactured, tested, and inspected in accordance with the specified requirements of referenced subsidiary specifications and standards and in accordance with this specification, as applicable.

4.3.2 Inspection of end item and preparation for delivery.

4.3.2.1 Examination of the end item. The end item shall be examined in accordance with the classification of defects listed for each of the applicable examinations.

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4.3.2.1.1 Examination of individual containers of ink for defects in appearance, design, and workmanship. An examination shall be made as indicated in table I. The sample unit for this examination shall be one filled glass bottle, polyethylene container or cylindrical tube, as applicable. The inspection and acceptable quality levels (AQL's) shall be as specified in 4.3.3 with the lot size expressed in units of individual containers of ink.

TABLE I. Visual examination (end item)

Examine	Defects
Appearance	Spillage of ink or leakage on outside of bottle or tube. Bottles or tubes not clean. Labels damaged, torn; not securely attached to bottle or tube.
Design	Container and closure not as specified.
Marking (bottle)	Not clear nor legible; not as specified.
Fill of individual container (except average contents)	Empty or broken bottle or tube; fill not as indicated.
Workmanship	Not as specified.

4.3.2.1.2 Examination of individual containers for average contents. The sample unit for this examination shall be one filled glass bottle, or polyethylene container or tube, as applicable. The inspection level shall be as specified in 4.3.3 with the lot size expressed in units of individual containers of ink. The average contents shall be not less than that specified. The bottle contents shall be measured to the nearest cc using an appropriate graduated cylinder. Type II, class 2 ink shall be reconstituted according to the manufacturer's directions before measuring.

4.3.2.1.3 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing, and marking requirements of section 5. Defects shall be scored in accordance with table II. For examination of contents, the sample unit shall be one shipping container prepared for delivery, selected just prior to the closing operations. Defects of closure specified shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-4 with an AQL of 4.0 percent defective.

TABLE II. Examination of preparation for delivery

Examine	Defects
Marking - (unit, intermediate, and exterior containers, as applicable)	Incorrect; omitted; illegible; improper size, location, sequence, or method of application; omission shelf life marking and other special markings, as specified.
Materials	Wrong type, class, or other category of required specifications. Material missing, damaged, or defective affecting serviceability.
Workmanship	Wrong or careless means in placement and use of labeling and packaging and packing materials such as loose or partially attached labels; flaps of containers incompletely closed; loose or over-tight straps; dangerous strap ends, protruding ails, and smeared adhesive.
Contents	Quantities of unit and intermediate containers not as specified. Wrong arrangement; bottle on side or upside down.

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TABLE II. Examination of preparation for delivery (con.)

Examine	Defects
Closure of shipping containers	Not as specified.
Gross weight of shipping container	In excess of specified weight. In excess of box limitation shown on boxmaker's certificate.

4.3.3 Inspection levels and AQL's. The inspection levels and AQL's for examination of the end item shall be as follows:

Specification reference	Inspection level	(Percent defective) AQL
4.3.2.1.1	S-4	2.5
4.3.2.1.2	S-2	-

4.4 Test procedure. Writing ink selected in accordance with 4.2.1 shall be tested as specified herein. Writing ink failing any test will be considered cause for rejection of the lot represented by the sample.

4.4.1 Writing tests. Clean and equip with a new sac a fountain pen which is known to be in good working order. The pen shall have a smooth medium point (commercially manufactured as a "medium point"). Allow the pen, after filling, to remain in a horizontal position for not less than 5 minutes with the cap removed. Make several lines of handwriting and figures on bond paper, type III, 25 percent rag, 40-pound basis weight, conforming to UU-P-121. The pen shall start writing when the point first touches the paper without undue pressure and without skipping or other impairment in the ink flow. The writing shall be free of feathered or ragged edges and shall not "strike through" the paper. The writing shall not smudge or offset onto an adjacent sheet of paper after the writing is made. Test this by simulating signing letters with a fountain or dip pen with medium point, writing a signature on five successive sheets of paper, allowing each sheet to fall on top of the previous sheet 5 seconds after the signature is finished.

4.4.1.1 Clogging. After the writing test described above, refill the fountain pen and replace the fountain pen cap. Keep the pen for 5 days in a horizontal position in a room or cabinet where the atmosphere is controlled at a temperature of 73° plus or minus 3.5°F. and relative humidity of 10 to 20 percent. At the conclusion of this period, repeat the writing test described above.

4.4.2 Stability. Place 100 milliliters of the sample ink in a container, cover the container with a watch glass to keep out dust, lint, etc., yet permitting access of air. Allow the ink to remain in the container for 15 days in a room at 73° plus or minus 3.5°F. and relative humidity of 45 to 55 percent. After this period of time, examine for scum or mold. The sample shall show no sediment when filtered through a Whatman No. 1 filter paper or equivalent.

4.4.3 Reaction to metal parts. Immerse for 1 week a common carbon steel pen point by a cotton thread in no less than 4 ounces of the sample ink. At least half of the pen point surface shall be immersed in the ink. At the end of the week, the pen point shall be in a serviceable condition so that it may be used for writing.

4.4.3.1 Stability after immersion. After the immersion test in 4.4.3, the writing ink shall still be in a serviceable condition. To determine this, subject 100 milliliters of this ink to the stability test in 4.4.2 for 8 days.

4.4.4 Washability. Cut pieces of white wool flannel and cotton broadcloth 1 inch square. Place approximately one drop of ink on each, and rub into the cloth. Allow to dry at room temperature for one-half hour. Rinse the pieces with water and place in a 100 milliliter beaker. Cover with a soap solution made by dissolving 0.5 grams of toilet soap, conforming to P-S-620, in 100 milliliters of distilled water. Cover the beaker with a watch glass and heat at 100°F. When the soap solution becomes discolored, pour it off and replace with fresh solution. Continue until no stain remains on the fabric. Time to remove stains shall not exceed 2 hours.

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4.4.5 Light fastness. Expose a sample of writing performed as specified in 4.4.1 in a carbon arc accelerated fading tester, such as the Atlas Fade-O-Meter, for 48 hours at a distance of 10 inches from the light source. Examine for legibility.

4.4.6 Fastness to water. Samples of writing performed as specified in 4.4.1 shall be allowed to dry for 2 hours and then immersed in distilled water for 24 hours. Remove the sheet and allow it to air-dry. Examine for legibility.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Ink shall be packaged level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Fluid ink. Fluid ink shall be furnished in 2-ounce, 4-ounce, 1-pint, and 1-quart commercial type bottles, as specified (see 6.2). No unit (individual) boxing is required for 2- or 4-ounce bottles. One-pint and 1-quart bottles shall be individually packaged in corrugated fiber boxes, double-slide type or equal, sized to snugly fit the bottle contained. Both ends of the box shall be tape-sealed to prevent accidental opening.

5.1.1.2 Ink powder. Powder of sufficient quantity to make 1 quart of fluid ink shall be packaged in sealed bottles or tubes in accordance with manufacturer's commercial practice.

5.1.1.3 Intermediate packaging. Twelve unit bottles of the 2- or 4-ounce size shall be intermediately packaged in a fiberboard box of appropriate size conforming to PPP-B-636, class domestic, grade 175 minimum, fitted with full- or shoulder-height corrugated fiberboard partition assembly providing separate cells or sections for each unit bottle. Box closures shall conform to the appendix to the box specification. No intermediate packaging is required for the 1-pint or 1-quart bottles packaged in unit boxes as specified in 5.1.1.1. Twelve or 24 unit packages of ink powder shall be packaged in a box conforming to PPP-B-566, PPP-B-636, PPP-B-665, or to PPP-B-676.

5.1.2 Level C. Fluid and powdered ink shall be packaged in accordance with manufacturer's commercial practice, providing that this will insure protection during shipment and safe delivery at destination.

5.2 Packing. Ink, packaged level A or C, shall be packed level A, B, or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Fluid ink. Four intermediate boxes (48 bottles) of the 2- or 4-ounce size; or six of the 1-pint or 1-quart size, shall be packed in a box conforming to PPP-B-636, V3c or V3cs. Box closure, waterproof sealing, and strapping shall conform to the appendix to the box specification, except that in lieu of steel strapping, boxes shall be reinforced with pressure-sensitive filament tape. Boxes shall be provided with caseliners conforming to MIL-L-10547, sealed in accordance with the appendix thereto.

5.2.1.2 Ink powder. Intermediate packages of powder shall be packed as specified for class 1 ink except that quantity of packages shall be such that the gross weight of boxes will not exceed 70 pounds.

5.2.2 Level B. Fluid and powdered ink shall be packed as specified for level A except that fiberboard boxes shall conform to class domestic, style RSC or CSSC, minimum grade 200. Closure shall be in accordance with method II of PPP-B-636.

5.2.3 Level C. Fluid and powdered ink, packaged as specified in 5.1, shall be packed to comply with the National Motor Freight Classification Rules or the Uniform Freight Classification Rules.

5.3 Marking. Marking shall be in accordance with 5.3.1 or 5.3.2, as specified (see 6.2).

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5.3.1 Military agencies. Marking shall be in accordance with MIL-STD-129.

5.3.2 Civil agencies. Marking shall be in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use. Writing inks covered by this specification may be used for ordinary office purposes or for permanent records.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type of ink required (see 1.2.1).
(c) Class and color if type II ink is ordered (see 1.2.1 and 1.2.2).
(d) Data, as required, regarding examination and tests; and inspection responsibility (for civil agencies only) if other than as specified herein (see 4.1).
(e) Size container desired for types I and II, class 1 ink (see 5.1.1.1).
(f) Levels of packaging and packing and any special marking for shipment required (see 5.1, 5.2, and 5.3).

COORDINATION WAIVED BY DOD

Preparing Activity:	User Interest:
GSA-FSS	Army - GL

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